Challenge Problems Week 2

Working with Qubits

UCQ Education Team

Every qubit can be written

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\varphi}\sin(\theta/2)\,|1\rangle$$

with $0 \leq \theta < \pi$, $0 \leq \varphi < 2\pi$. An explicit mapping from qubits to the Bloch sphere is given by

$$\cos(\theta/2)\,|0\rangle + e^{i\varphi}\sin(\theta/2)\,|1\rangle \mapsto (\theta, \varphi)$$

in spherical coordinates where φ is the azimuthal angle (the "physics" convention).

Problem 1. Given any two states $|\psi_1\rangle$ and $|\psi_2\rangle$, show that there is an set of possible measurements (orthonormal basis) $\{|b_1\rangle, |b_2\rangle\}$ such that

$$|\,\langle b_1|\psi_1\rangle\,|^2 = |\,\langle b_2|\psi_1\rangle\,|^2 = |\,\langle b_1|\psi_2\rangle\,|^2 = |\,\langle b_2|\psi_2\rangle\,|^2 = \frac{1}{2}$$

i.e. both states have a 50-50 chance of measuring $|b_1\rangle$ or $|b_2\rangle$. What is the geometric meaning of this basis on the Bloch sphere? (This was used by Gupta et al. (2025) as part of an algorithm for fast quantum state certification.)

Problem 2. Show that there are valid quantum states

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle \qquad\qquad |a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

that cannot be written as the tensor product of two qubits. These are called *entangled states*.